

Bosen Yang

3130 Sawtelle Blvd, Los Angeles, CA, 90024 | (424)-514-9653 | bosen@umich.edu | [LinkedIn](#) | [Webpage](#)

EDUCATION

- University of California, Los Angeles** (Cum Laude, 3.936) Degree Conferred – Jun 2026
- *B.S. in Applied Mathematics (Computing Specialization), B.S. in Statistics and Data Science*
 - Core Coursework: Real/Complex Analysis, Numerical Analysis, Math Modeling, Linear Algebra, Diff Eqn, Graph Theory, OOP, Software Tools, Data Structure, Prob Theory, Comp/Math Stats, Linear Model, Exp Design, Data Mining, Optimization, Monte Carlo, Consulting.
- University of Michigan, Ann Arbor** Degree Expected – Dec 2027
- *M.S. in Applied Statistics*

EXPERIENCE

- Research Assistant** *UCLA Trustworthy AI Lab – Directed by Professor Cheng* Los Angeles, CA; Apr 2026 – Present
- Simulated **interactive benchmark games for LLM agents** such as prisoner's dilemma with communication in Python. Conducted analysis of multi-agent trust-building, betrayal, decision-making with and without conversation from Stats 263 class game response data.
- Directed Reading** *UCLA Mathematics – Mentored by PhD Candidate William Chang* Los Angeles, CA; Jan 2026 – Present
- Investigated theoretical foundations of online learning for sign-activation neural networks.
 - Investigated margin-preserving depth reduction to eliminate depth dependence in existing neural network mistake bounds. Analyzed polyhedral representations of threshold networks and established margin preservation for collapsed two-layer architectures.
- Directed Reading** *UCLA Stats 199 – Mentored by Professor Michailidis* Los Angeles, CA; Apr 2026 – June 2026
- One-on-one simulation study on **high-dimensional statistical estimation under network dependence** in Bayesian perspective.
 - Implemented data simulation with Gibb's sampling and evaluated estimation performance under different priors in Python.
- Data Science / ML Intern** *Jietai Solar* Anhui, Chuzhou, CN; June 2024 – Aug 2024
- Analyzed photovoltaic product sales and production cost data for systematic manufacturing schedule optimization. Adopted time-series forecasting and regression modeling techniques in R and Python to model short to medium term demand.
 - Queried and imported historical production volume and profit/loss data using SQL, and constructed forecasting frameworks informed by cost accounting and market trends.
 - Implemented Machine Learning algorithms (RFR, LSTM etc.) with mlr3 and scikit-learn. Performed cross validation and scenario analysis to improve forecast stability and accuracy on rolling basis.
- Instructional Assistant & Reader / Grader** *UC San Diego Mathematics* San Diego, CA; Sep 2023 – Dec 2023
- Course: MATH 20E Vector Calculus - Holding 4 discussion sections and weekly office hours; grading homework/exams; proctoring exams.

PROJECTS

- Multi-Sensor Temperature - Enzyme Activity Experiment & Data Denoising** July – Aug 2025
- Designed and built a multi-sensor temperature monitoring module using Arduino UNO controller to study ALP enzyme solution.
 - Simulated heat diffusion via the fundamental solution of the heat equation, yielding 3D Gaussian heat distribution. Designed a monotonic heating experiment with delayed enzyme reaction modeling and applied method of char. to approximate optimal ALP activity temperature.
 - Designed an integrated real-time denoising framework combining Dynamic Inverse Variance Weighting, FFT, and Kalman Filter in R. monitoring and control within the designated interval of the module.
- International Global Terms of Trade Time Series Forecasting** Aug – Sep 2025
- Analyzed Terms of Trade (ToT) time series (CN, JP, IN, AU, US) and constructed multivariate forecasting models in R
 - Implemented and compared ARIMA, ETS (with Denton-Cholette Method), Multivariate, and GARCH model using forecast, vars, and rugarch in R, with stationarity and normality tests, Box-cox transformation and correlation analysis to inform model selection.
 - Conducted trend analysis of ToT and derived econometric insights to inform business decision-making and macroeconomic forecasting.

SELECTED REVIEWS

- [1] Yang, Bosen. "Application of matrix decomposition in machine learning." *2021 IEEE International Conference on Computer Science, Electronic Information Engineering and Intelligent Control Technology (CEI)*. IEEE, 2021.
- [2] Yang, Bosen. "Internal Combustion Engine Vehicles Fuel Efficiency Regression Analysis with Physical and Engineering Perspective." *Highlights in Business, Economics and Management* 65 (2025): 382-392.

SKILLS

- **Programming:** R, Python, MATLAB, Java
- **Software Tools & Frameworks:** LaTeX, SQL, VS Code, Git, Word/Excel